

NDTwin Network Digital Twin

The Manual, Proven End to End

— and what one unstated build flag is worth

OUTLINE

- 1 The manual** pp. 4–6
installed from the public document, in a clean room
- 2 What the experiments say** pp. 7–17
one build flag = 8× · capacity depends on flow count · 18 papers
- 3 Engineering, and what's next** pp. 18–23
the traffic round · a criterion that measured the wrong thing

WHERE WE LEFT OFF

three written on 27 August · one asked for since

THE ASK	TEST SET IN ADVANCE		TODAY
THE MANUAL	install from the public document, clean room	ANSWERED	§1 · and the site is open in the next tab
DOES MERGE MOVE THE WALL	the 1-in-8 cell's loss moves, or it does not	ANSWERED	72 cells ran — the criterion could not see the wall (p. 19)
FAST BUILD DEFAULT	the L0–L4 suite passes on the fast build	IN PART	install and provenance verified · the suite has not run
THE BOOT RING	arms to 0/6–0/8 on the fixed build	NOT RUN	deliberately deprioritised · carried forward as written

SECTION 1

The manual

Installed from the public document in a clean room — the site itself is the demo

IT INSTALLS, AND IT RUNS

clean-room VM • every command as printed

31

installation steps run verbatim — zero non-zero exit codes

12/12

hosts reach each other on the fabric the manual builds

29/29

documented API endpoints exist, with the right methods

7

checks pass on the optional fast build — the debug install untouched

16 s

for the whole stack to converge, on a real clock

12

routes with no page — 29% of the HTTP surface

→ [the manual itself, live](#)

WHAT RUNNING IT FOUND

defects the document could not show • four fixed, two open

THE MANUAL

grep for the wrong error string • an undocumented interactive prompt • “wait 60 s” measured 80

THE PROGRAM

a second proxy wrote rules before noticing the port was taken • the lock API accepted malformed requests

A GREEN CHECK

“model matches fabric: 4 hosts” printed over 128 — it compares the model with the model

A PHANTOM ROW

a queued, unprogrammed request served as a flow-table row • visible at $t = 0$ only, on both planes

STILL OPEN

the Ubuntu Server variant • twelve undocumented routes, one of them called by the shipping GUI

The last two are system defects — found by running the document, not by reading it.

SECTION 2

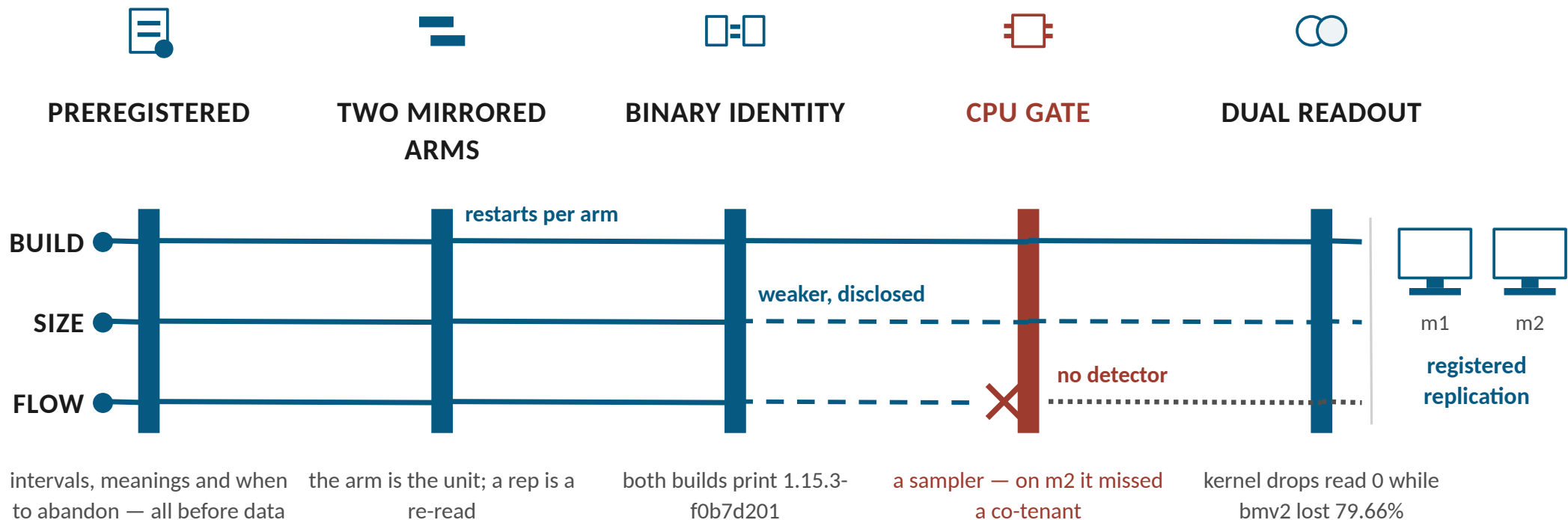
What the experiments say

One unstated build flag, a capacity that depends on flow count, and eighteen papers that state neither

HOW WE MEASURED

five gates · three experiments · and where they do not hold

ENTRY $\times 1.5$ rungs · clean = loss $\leq 0.5\%$ on 3-rep medians · ± 1 rung is the resolution



Not “we were rigorous” — this is the shape of it, and where it breaks.

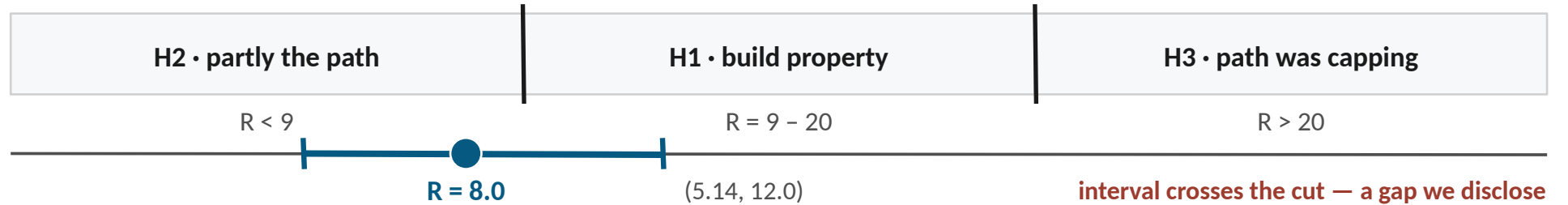
Transcribed from paper/abstract §“Three preregistered measurements”. Solid = as registered · dashed = weaker, disclosed · broken = the gate did not exist for that experiment.

EVERY OUTCOME HAD A MEANING FIRST

three variables • three registered bands each • no “failed experiment”

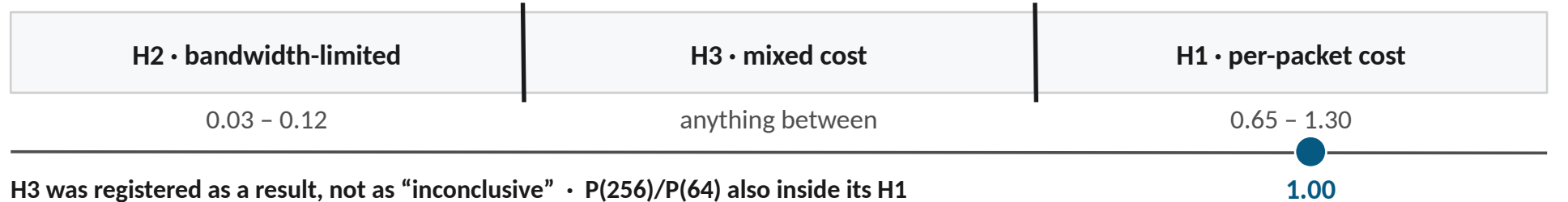
BUILD

ratio $R = \text{fast} \div \text{stock}$,
one hop



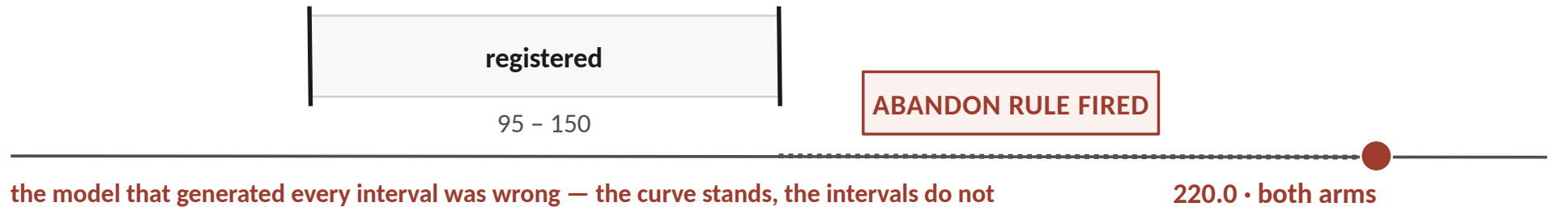
SIZE

ratio $P(1024) \div P(64)$,
delivered pps



FLows

aggregate at $n = 2$,
Mbit

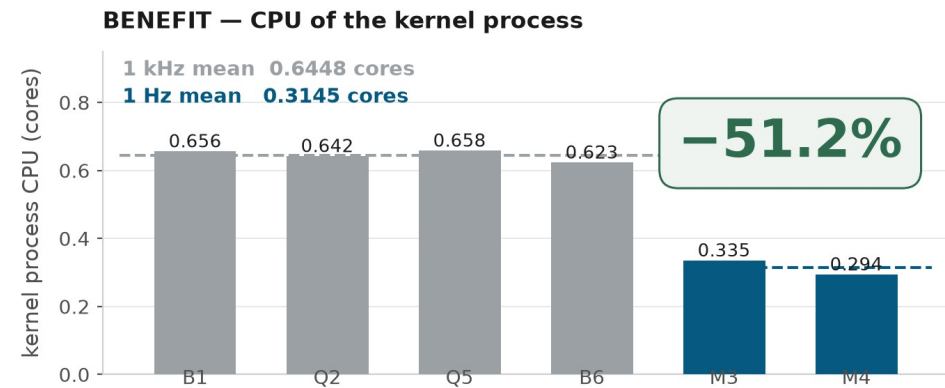
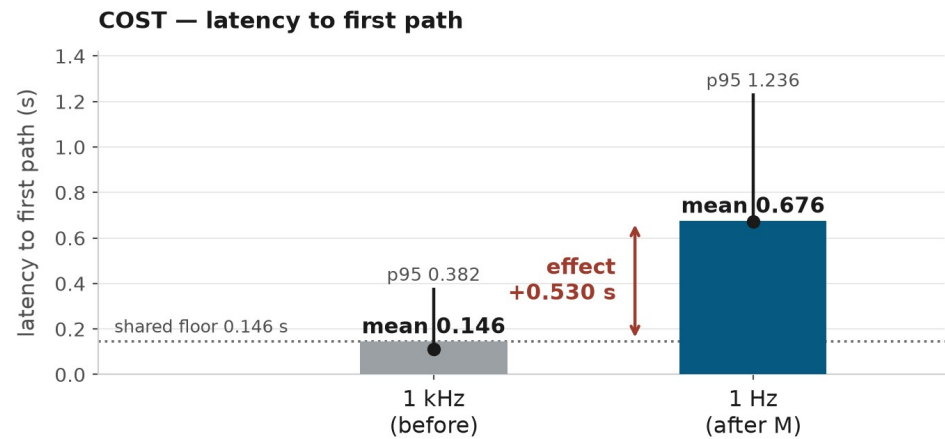


A result is which band it landed in. All three bands were results.

Bands drawn equal-width on purpose: before the data, all three were equally admissible. Intervals transcribed from the three PREREGs; the abandonment rule is 48487ed, written before data.

Ticket M, both halves measured: it costs half a second and buys half a core

six mirrored arms • one fabric generation • 1 kHz vs 1 Hz sampling



COST PRE-REGISTERED

effect +0.530 s, inside the registered 0.4–0.6 s band

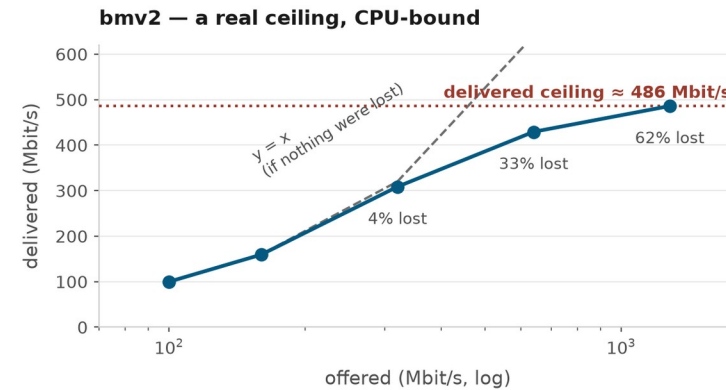
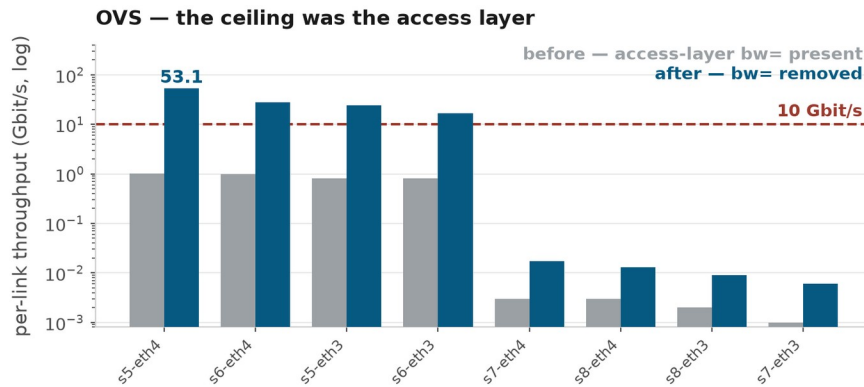
BENEFIT POST-HOC

not registered before the data was read — an observation, not a result

The ceiling was the access layer; BMv2's is real

A working point from one plane means nothing on the other

left: OVS core links, access-layer bw= removed, 32 TCP flows, 53.1 is one link's ECMP share · right: bmv2 —O3 build, single flow



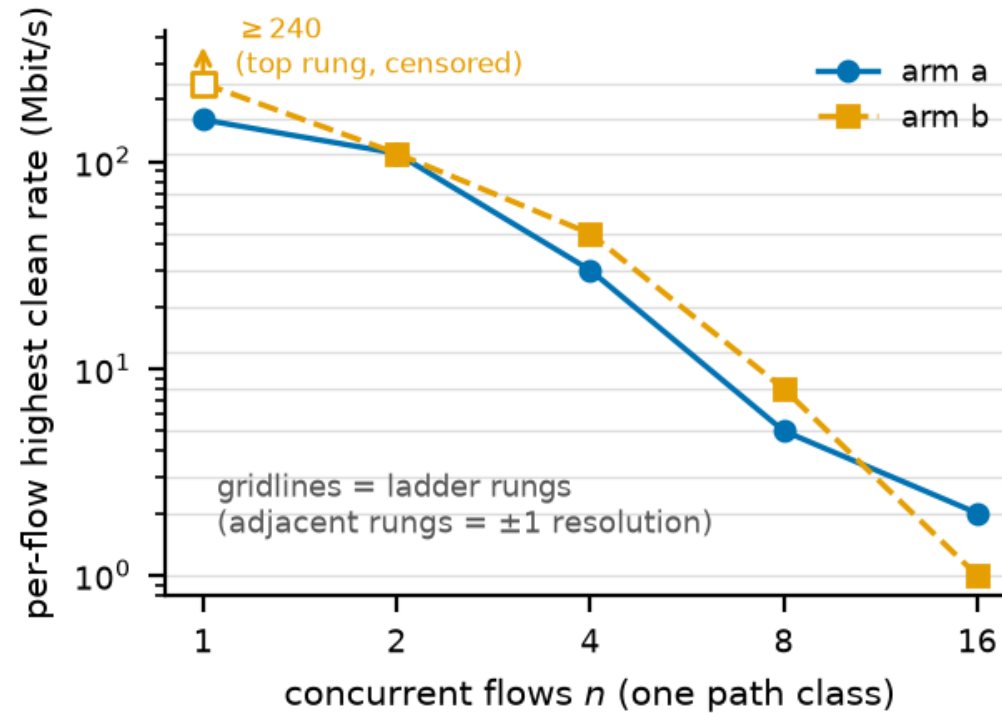
THE FOUR SHORT BARS

ECMP hashed onto the top four — not a capacity floor

WHAT 53.1 IS

an interface-counter rate · 32 TCP flows · one link's ECMP share · host CPU 98.7%

Capacity is a function of flow count



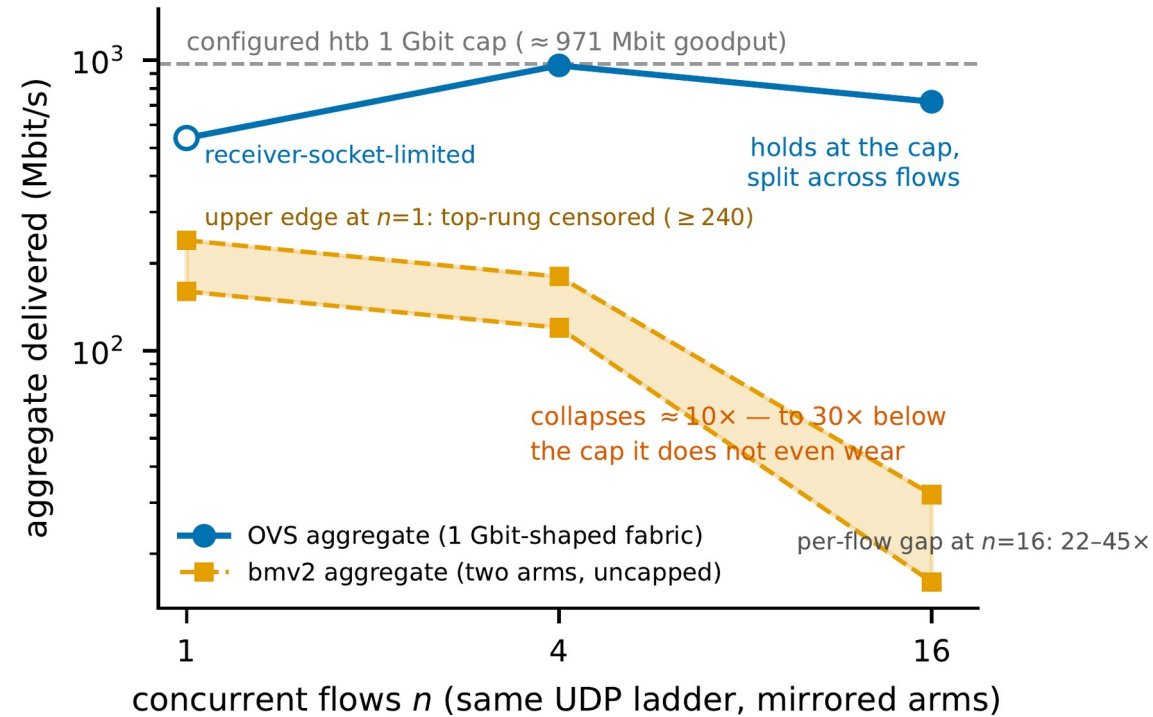
UDP

the ladder is iperf3 -u and “clean rate” is read from loss — this is a UDP claim

THE CONTROL

same path on TCP: $T(16)/T(1) = 1.222$ and 1.108 — no collapse

One plane holds its cap; the other collapses below it



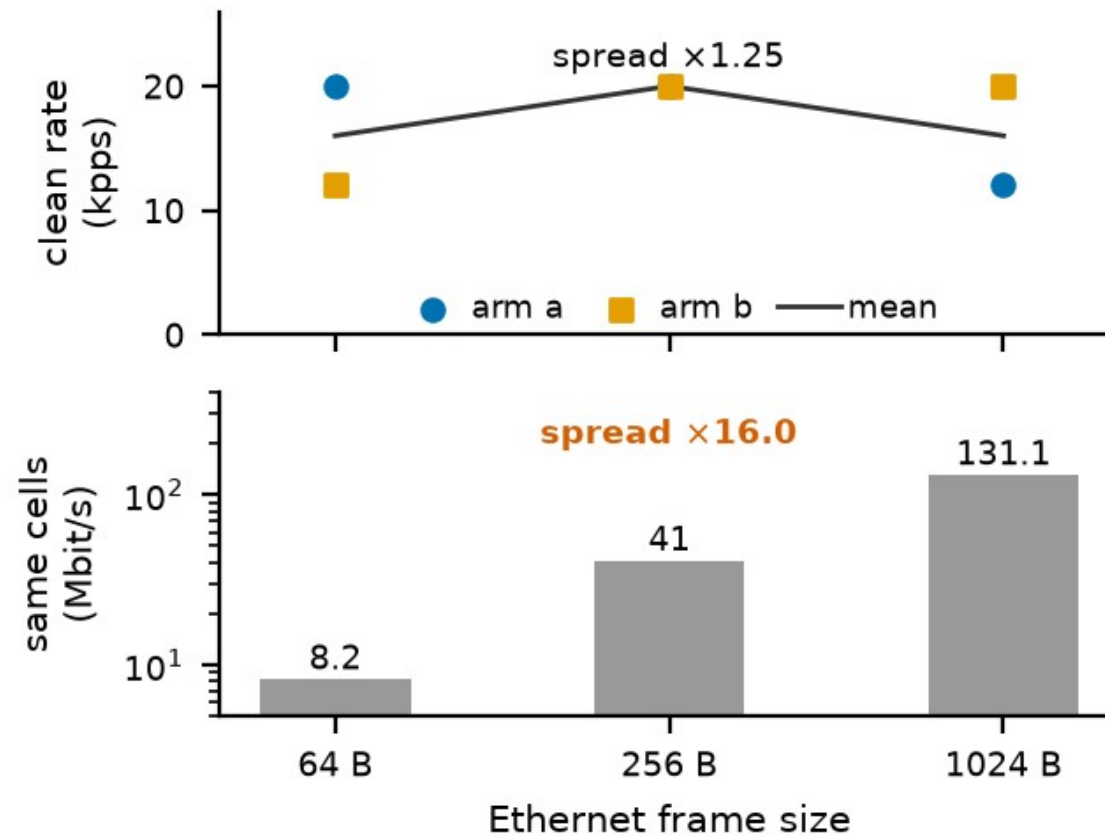
UDP, BOTH PLANES

both ladders are `iperf3 -u -l 1400` · with TCP the collapse does not appear

SO THE CLAIM IS

“UDP collapses $\approx 10\times$ ”, not “BMv2 collapses”

Same data, two units, sixteen times apart



THE AMBIGUITY

packet rate moves 1.25× while bit rate moves 16× — Mbit/s without a packet size is a 16× hole

18 PAPERS, AND WHAT THEY REPORT

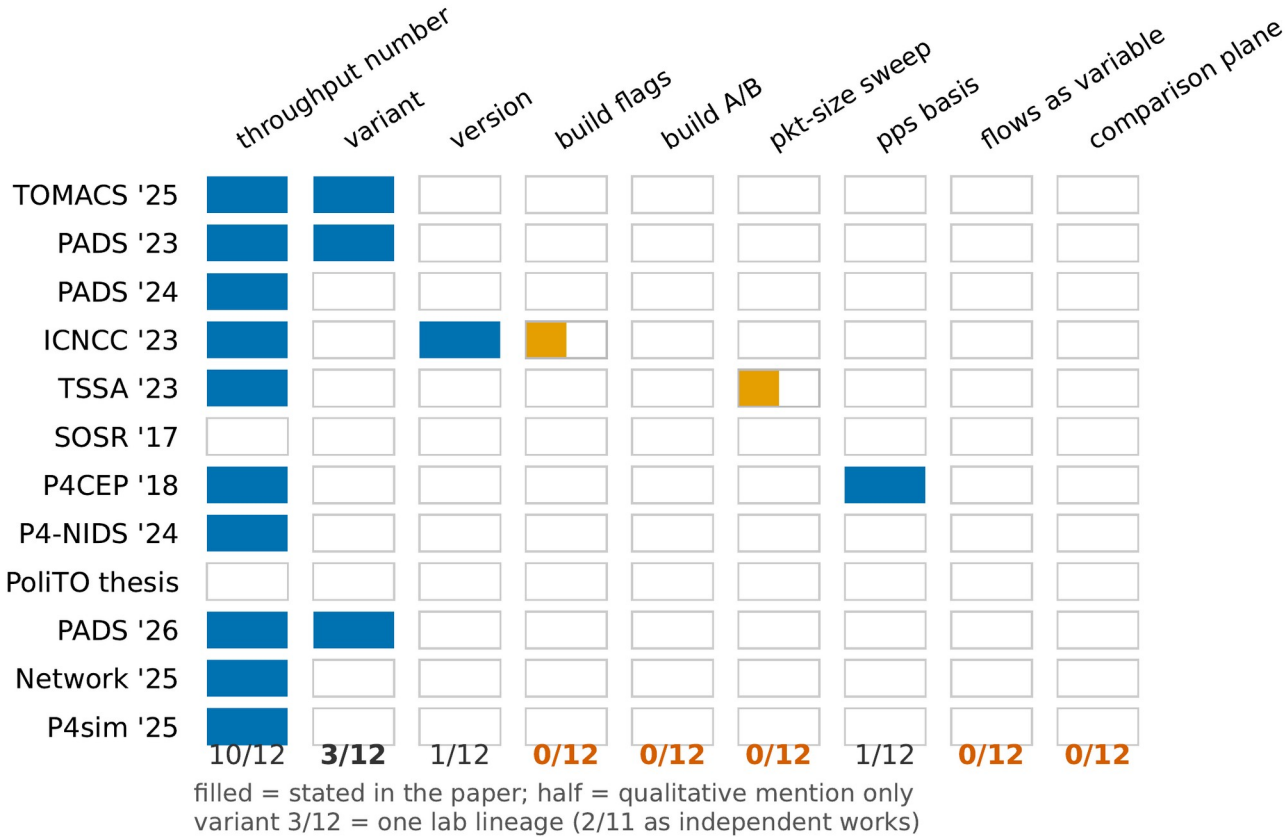
12 of them measure BMv2 • counted, not estimated

reporting dimension	of the 12	note
reports a throughput number	10 / 12	two measure latency only
states build flags	0 / 12	the variable worth 8× here
runs a build comparison	0 / 12	
names the BMv2 variant	3 / 12	all from one lab lineage
states the version	1 / 12	
sweeps packet size	0 / 12	
uses flow count as a variable	0 / 12	

THE GAP

the one variable we measured at 8× is the one none of them state — in our sample of 18

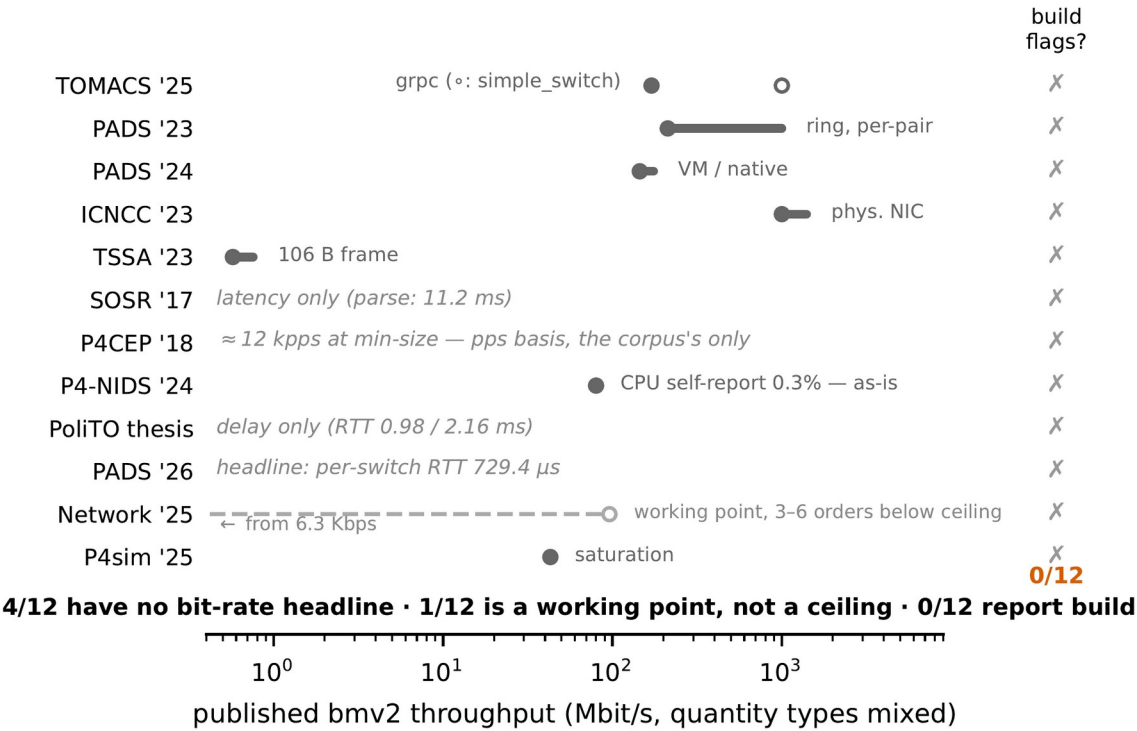
Twelve papers, nine reporting dimensions



THE GAPS ARE THE DATA

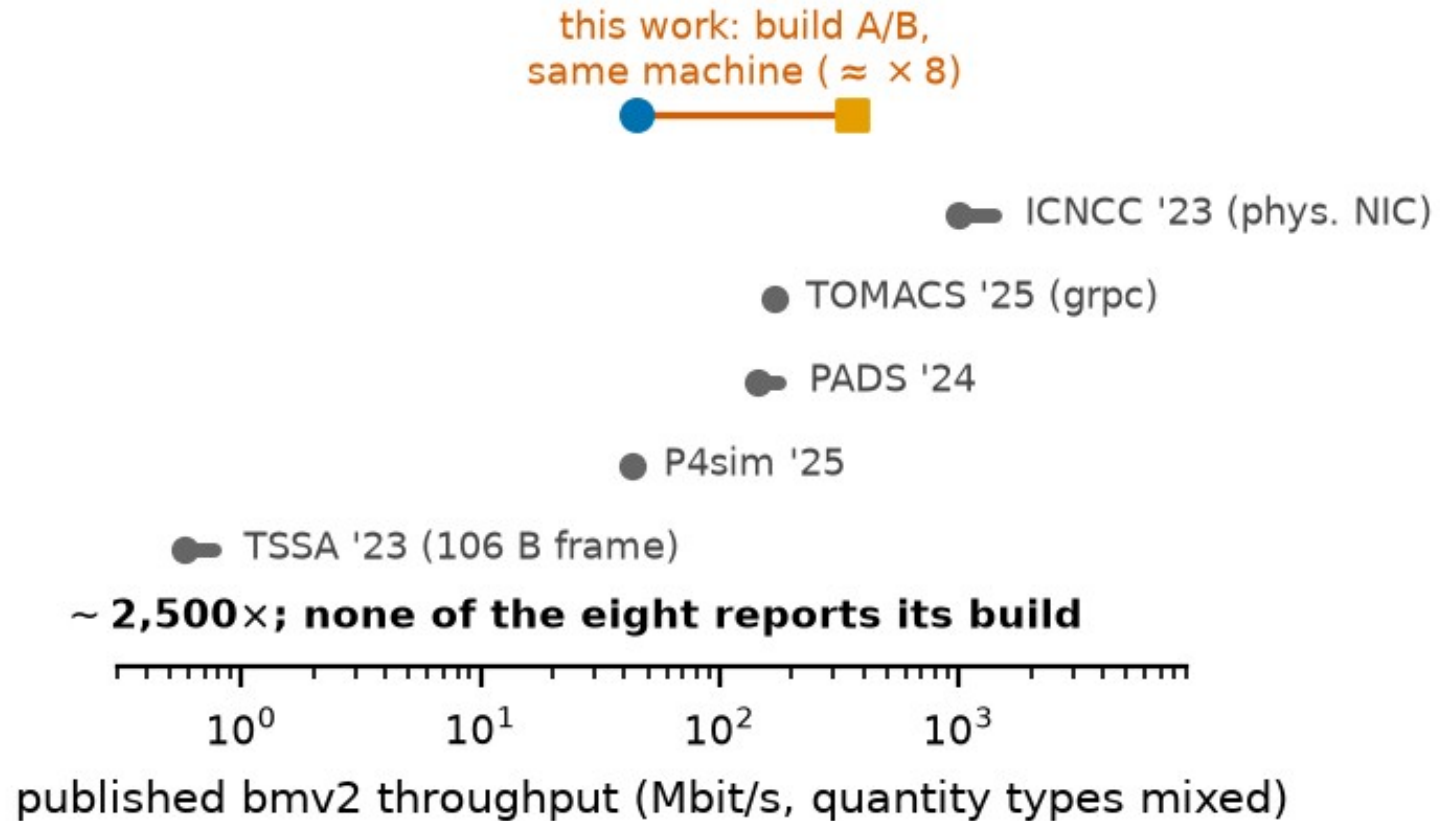
column totals are asserted against the survey table — a drift crashes the plot rather than drawing it

Twelve headline numbers, one axis



A THIRD CANNOT BE PLACED	4/12 publish no bit-rate headline • 1/12 is a working point, not a ceiling
NOTHING WAS CONVERTED	a packets-per-second headline stays in packets per second

Self-reported throughputs span ~2,500×



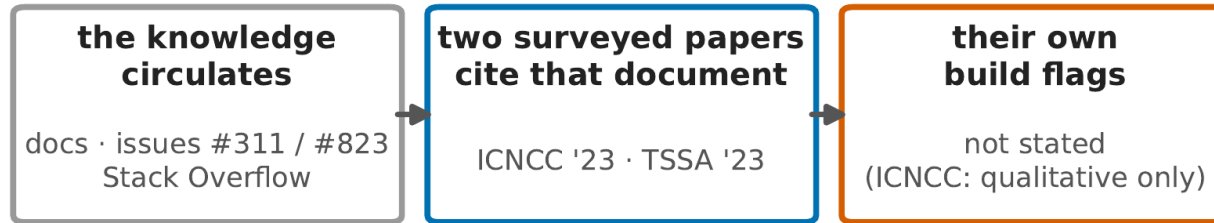
FOR SCALE

~0.57 Mbit/s to ~1.4 Gbit/s • one build flag on one machine is worth 8× of that

Known is not the same as reported

*“which flags were used to build bmv2:
this can have a massive impact.”*

— p4lang/behavioral-model, docs/performance.md, which also gives recommended flags
and a reference figure (~1,047 Mbps median / 80 kpps, c4.2xlarge)



Common knowledge is not a reporting norm — build flags stated: 0/12.

contrast: Zhang, Comput. Netw. '21 (7 software switches, no bmv2) names a commit per switch
(FastClick 9d5e9c6, t4p4s b1161b2, ...) + per-switch parameters — the norm exists next door

DESK CHECK

sources read, not re-measured • BMv2's own page says the debug build is slow, and two papers cite it

AND STILL

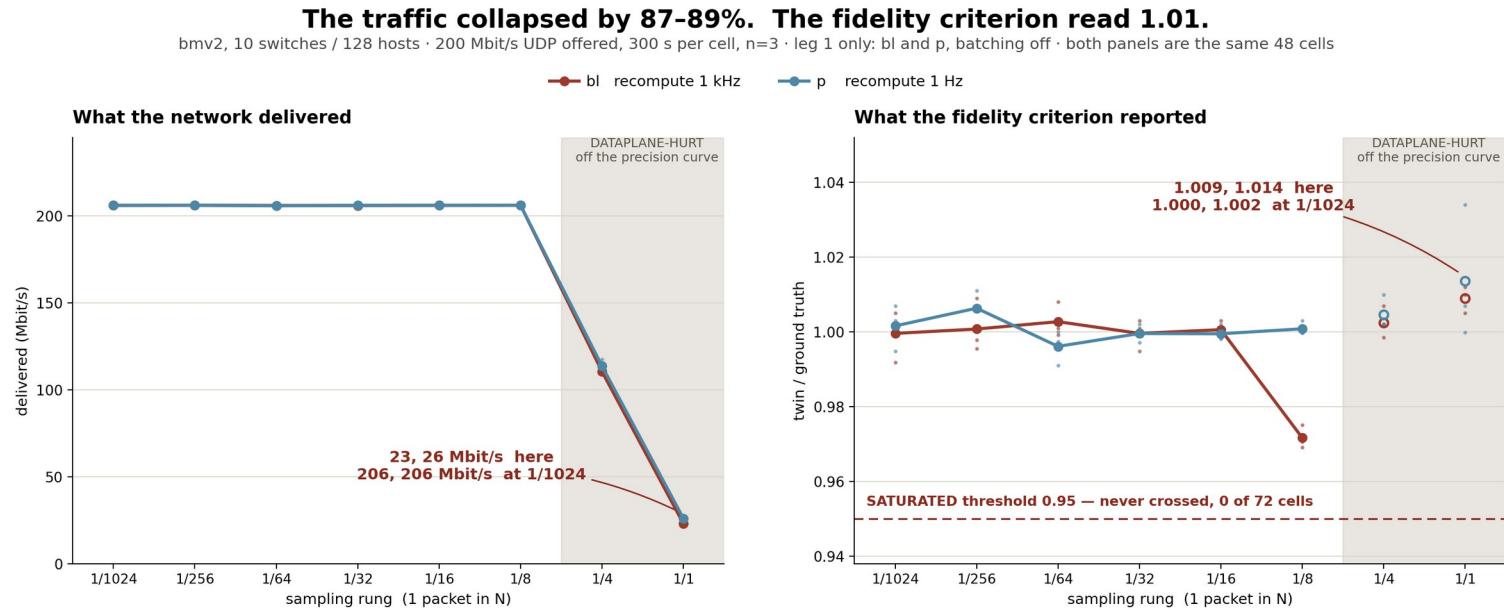
neither states its own build • build flags stated: 0/12

SECTION 3

Engineering, and what's next

The traffic round, a criterion that measured the wrong quantity, and what the next report will carry

The criterion never measured the quantity its name says



NOT READ OUT

SATURATED fired 0 times in 72 cells · all four arms right-censored · indistinguishable, not equal

THE FLAT LINE MISLEADS

twin ÷ truth reads 1.009 at the rung that destroys 87-89% of the traffic

BATCHING NOT ESTIMABLE

batch on/off is aliased with leg and time of day — a confound, not noise

THE TRAFFIC ROUND

six arms • registered before any data

IT DISCRIMINATES

the flow endpoint went 8 → 12 under load and tracked the churn — the gate held on both arms

CONTRACT

39/39 against the live kernel, plus a 52-check self-test

THE 128-HOST EXAMPLE

the manual's own worked example passes as printed

THE HALF NOT ASKED

the timing question is unanswerable at 128 hosts — one sample costs 735 ms and needs 500

AND NOT BY SAMPLING HARDER

the instrument is slower than the thing it measures • answered at 4 hosts instead

ONE FLAG, TWO MEANINGS

a decision and an observation, sharing one bit

WHO WRITES `is_up`

THE POWER API

a decision — I commanded it off

THE 1 Hz LIVENESS WORKER

an observation — it answers

One bit cannot tell “the wall switch is off” from “the bulb burned out”.

RULED

split into `admin_state` (the command) and `reachable` (the probe) • `is_up` stays as an alias for `reachable`

WHAT IT COST, MEASURED

8–13 s

of “on” after the machine was commanded off — the liveness cache writes it back within 0.3–1.5 s. 18/18 across both arms

OFF vs ON

two switches equally dead: the commanded one reads OFF, the crashed one reads ON. The endpoint reports the order, not the state

never fires

the three-state check for “something died that should not have” reads the same bit on both sides

Decided, not shipped: the fix is dispatched and not yet on trunk.

WHAT THE NIGHT ROUND FOUND

six endpoints • what they report vs what the switch does

A LINK COMES BACK

a link declared down through the API is silently restored by the 30 s topology poll — 21.5–25.4 s. A real cut never restores itself

SHUTDOWN KEEPS WRITING

“All subsystems stopped. Exiting.” is printed, and 10.7 s of southbound writes follow it — no log line mentions them

A COUNTER COUNTS POSTS

20 installs of one match: succeeded +20, the switch +1 • 15 deletes of a match that is not there: succeeded +15, the switch +0

A 200 THAT INSTALLED NOTHING

two reserved group ids answer “200 installed” and never appear in groupdesc • the delete branch of the same code answers 404 honestly

A METRIC THAT NEVER MOVES

path switch_count is computed from the static model: identical across 12 paths with 8 of 40 switch-to-switch edges down

40× FOR THE SAME WRITE

meter install 1024 ms against group install 20 ms — both are one OpenFlow write, measured at the client

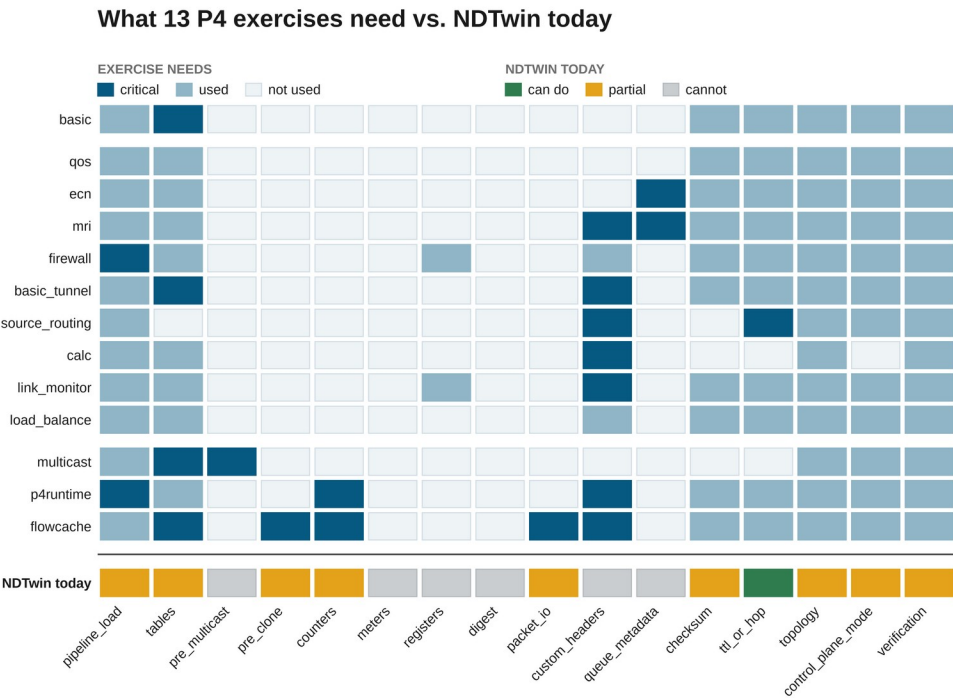
Current behaviour only — the fixes were still running when these were taken.

Kernel cca5e3e4, helper 6685d3a9; trunk 4088b237 / f943de8f by round. Figures and raw logs in figures/overnight-0904/. The shutdown baseline (~2.4 s idle) is read from another round, on another binary, and is marked as such.

What the tutorials ask for

13 exercises • 16 capability dimensions • read, not run

READ-CODE / NOT RUN



PIPELINE LOAD, NOT FEATURES

EXACT + LPM ONLY • 1/13 UNCHANGED

8 of the 13 want one thing: load the .p4 they ship with • 6 of the 16 dimensions only ask whether NDTwin's own program has a feature

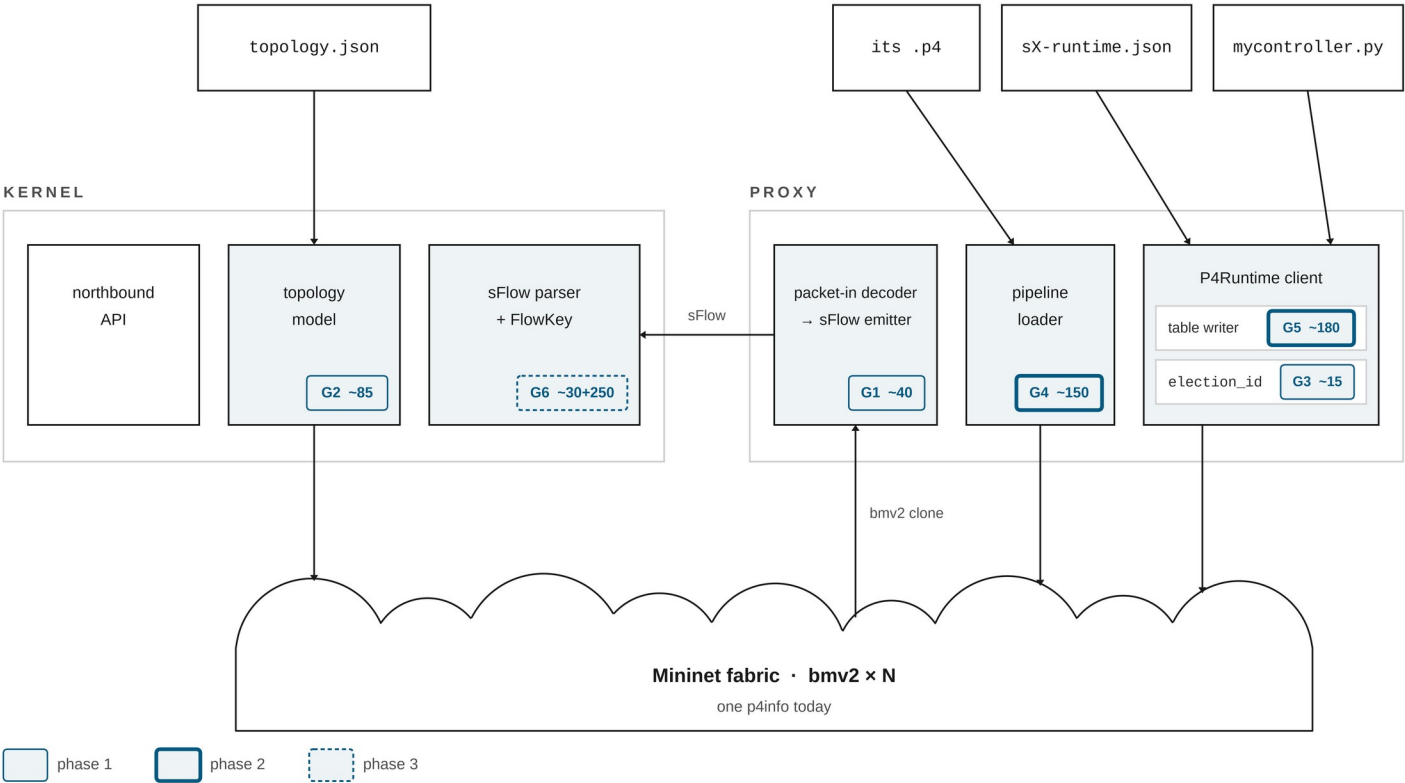
0/13 need meters or digest • only basic runs as-is, and even then the skeleton punts where the exercise drops

Where the five gaps live

proxy side and kernel side • sizes are estimates

READ-CODE / NOT RUN

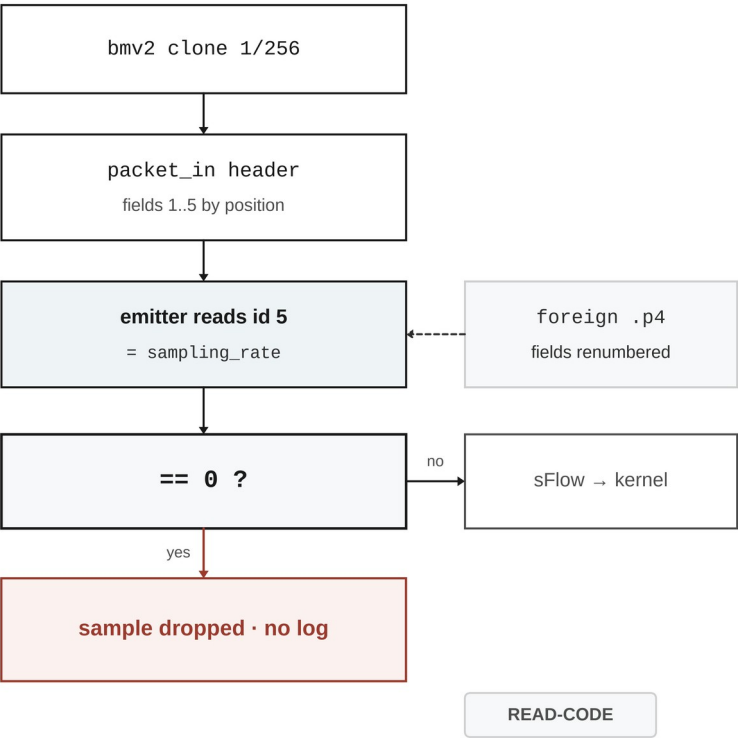
EXERCISE



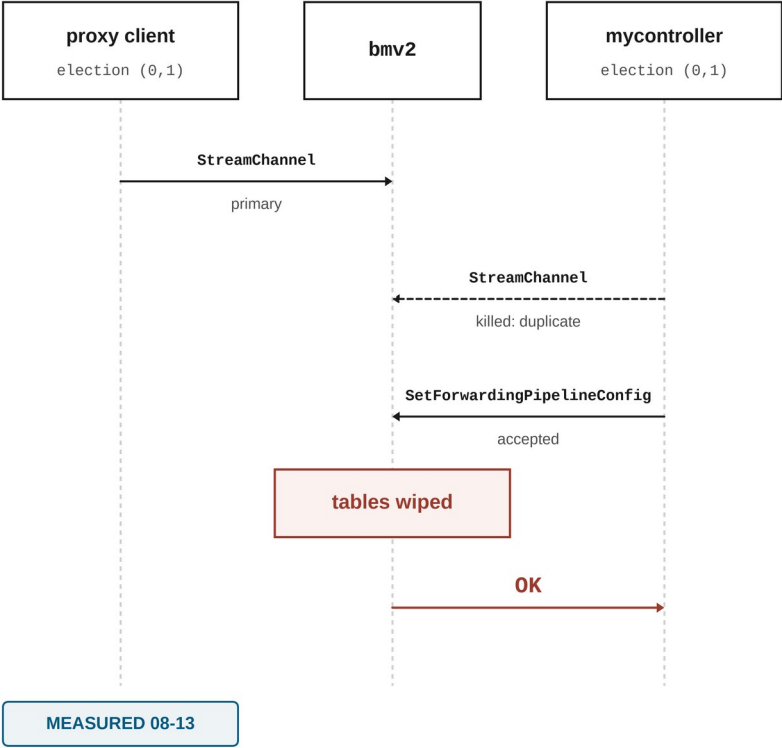
NOT ONE LINE WRITTEN the badges are estimated sizes, not progress • border style is the phase, and phase three is the sFlow parser

WHAT ONLY A FOREIGN P4 PROGRAM REVEALS

the same p4info on ten switches, for months



① fields read by position

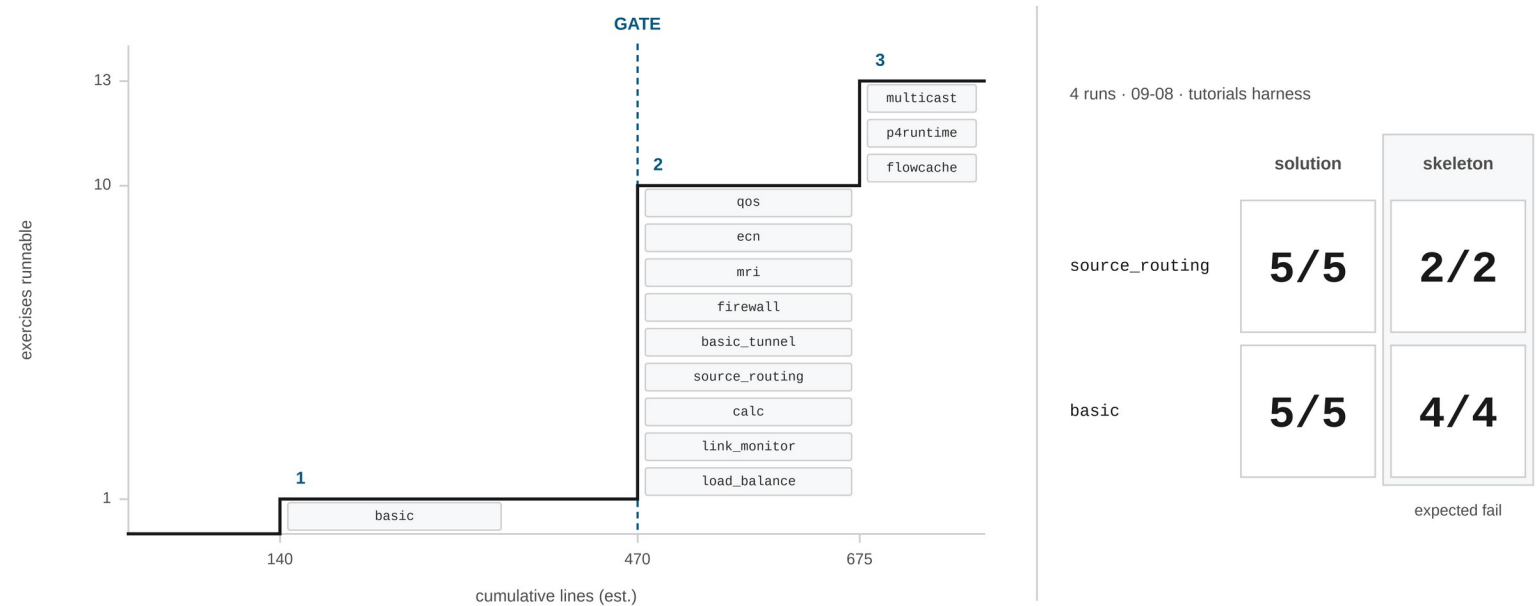


② two controllers, one switch

① line-checked at `sflow_emitter.py:425-429,468-470` and `p4_client.py:217,222` — the mechanism, not an observation. ② measured 2026-08-13; bmv2 follows the spec throughout.

Three phases to a regression gate

~675 lines • the gate opens after phase two



0 OF 13 ON NDTWIN

WHY THE SKELETONS

the 2×2 ran on the tutorials' own harness with their switch binary — it shows the exercises behave as expected, not that NDTwin can host them

an expected failure, observed • a gate nobody has seen go red is not a gate

WORTH WRITING UP?

asking, not arguing

QUESTION

Has anyone quantified what an unstated build flag costs?

MEASURED

one build flag • 8.0× on a single hop • 12× across three

MEASURED

unit ambiguity adds 16× on the same data

SURVEYED

0/12 state that flag, in our sample of 18

DESK CHECK

so a ~2,500× spread has no attributable cause — from reading, not measured

THE QUESTION

we did not find anyone who has done this • a pointer would be the useful answer

Not that this is unknown — that being known has not made it reported.

PLANNED FOR THE NEXT REPORT

three, each with the test that settles it

1

Measure the ceiling with a criterion that can see it

SATURATED never fired in 72 cells, so the arms cannot be told apart.

Done when a registered criterion reads out a rung the data plane actually fails at.

2

Finish the third question on its own terms

The fast build's install and provenance are verified. The functional suite is not.

Done when the whole L0-L4 suite passes on the fast build.

3

Close the manual line to publishable grade

Install, whole-stack and API page are green. Four executable pages remain.

Done when every executable page has been executed.

WHERE IT STANDS

settled · open

SETTLED

The manual installs	clean room · 12/12 up
The API page	29/29 correct
One build flag	8.0× · 12× across three hops
Units	1.25× or 16×, by choice of unit
Capacity	a function of flow count (UDP)
The survey	0/12 state build flags

OPEN

The sampling ceiling	measured fidelity, not the wall
Fast build as default	L0–L4 has not run
Manual, remaining	four executable pages
The boot ring	arms never taken to 6–8
Batching	aliased with time of day
Undocumented routes	twelve · one used by the GUI

Marked items are what the next report answers.